

```

(kali㉿kali)-[~]
$ mkdir -p ~/CIP-B103-Lab8/{evidence,working,exported,reports,screenshots,scripts}

(kali㉿kali)-[~]
$ cd ~/CIP-B103-Lab8

(kali㉿kali)-[~/CIP-B103-Lab8]
$ pwd
/home/kali/CIP-B103-Lab8

(kali㉿kali)-[~/CIP-B103-Lab8]
$ find . -maxdepth 1 -type d -print
.
./working
./screenshots
./evidence
./scripts
./reports
./exported

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo systemctl enable --now apache2
[sudo] password for kali:

```

```

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-08-10 18:53:23 EDT; 2min 31s ago
 Invocation: 2a1cbd563ff246c6a843f21a71f0836b
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 892 (apache2)
  Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 6 (limit: 2075)
   Memory: 35.7M (peak: 36.7M)
      CPU: 546ms
   CGroup: /system.slice/apache2.service
           └─892 /usr/sbin/apache2 -k start -DFOREGROUND
             └─950 /usr/sbin/apache2 -k start -DFOREGROUND
               └─951 /usr/sbin/apache2 -k start -DFOREGROUND
                 └─952 /usr/sbin/apache2 -k start -DFOREGROUND
                   └─953 /usr/sbin/apache2 -k start -DFOREGROUND
                     └─954 /usr/sbin/apache2 -k start -DFOREGROUND

Aug 10 18:53:21 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 10 18:53:23 kali apachectl[892]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name: []; please see the README file for the appropriate httpd.conf and httpd-ssl.conf settings.
Aug 10 18:53:23 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
Hint: Some lines were ellipsized, use -l to show in full.

```

```

(kali㉿kali)-[~/CIP-B103-Lab8]
$ curl http://127.0.0.1

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Debian Default Page: It works</title>
    <style type="text/css" media="screen">
      * {
        margin: 0px 0px 0px 0px;
        padding: 0px 0px 0px 0px;
      }

      body, html {
        padding: 3px 3px 3px 3px;

        background-color: #D8DBE2;

        font-family: Verdana, sans-serif;
        font-size: 11pt;
        text-align: center;
      }

```

```
div.main_page {
  position: relative;
  display: table;

  width: 800px;

  margin-bottom: 3px;
  margin-left: auto;
  margin-right: auto;
  padding: 0px 0px 0px 0px;

  border-width: 2px;
  border-color: #212738;
  border-style: solid;

  background-color: #FFFFFF;

  text-align: center;
}

div.page_header {
  height: 99px;
  width: 100%;
```

```
background-color: #F5F6F7;
}

div.page_header span {
margin: 15px 0px 0px 50px;

font-size: 180%;
font-weight: bold;
}

div.page_header img {
margin: 3px 0px 0px 40px;

border: 0px 0px 0px;
}

div.table_of_contents {
clear: left;

min-width: 200px;

margin: 3px 3px 3px 3px;

background-color: #FFFFFF;
```

```
}  
  
div.table_of_contents_item {  
    clear: left;  
  
    width: 100%;  
  
    margin: 4px 0px 0px 0px;  
  
    background-color: #FFFFFF;  
  
    color: #000000;  
    text-align: left;  
}  
  
div.table_of_contents_item a {  
    margin: 6px 0px 0px 6px;  
}  
  
div.content_section {  
    margin: 3px 3px 3px 3px;  
  
    background-color: #FFFFFF;  
  
    text-align: left;
```

```
div.content_section_text {
  padding: 4px 8px 4px 8px;
  color: #000000;
  font-size: 100%;
}

div.content_section_text pre {
  margin: 8px 0px 8px 0px;
  padding: 8px 8px 8px 8px;

  border-width: 1px;
  border-style: dotted;
  border-color: #000000;

  background-color: #F5F6F7;

  font-style: italic;
}

div.content_section_text p {
  margin-bottom: 6px;
}

div.content_section_text ul, div.content_section_text li {
```

```
div.section_header {
    padding: 3px 6px 3px 6px;

    background-color: #8E9CB2;

    color: #FFFFFF;
    font-weight: bold;
    font-size: 112%;
    text-align: center;
}

div.section_header_red {
    background-color: #CD214F;
}

div.section_header_grey {
    background-color: #9F9386;
}

.floating_element {
    position: relative;
    float: left;
}
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ printf '<!DOCTYPE html>\n<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: Fuseini Imoru Kantuogaa</p></body></html>\n' | sudo tee /var/www/html/index.html
<!DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: Fuseini Imoru Kantuogaa</p></body></html>

(kali㉿kali)-[~/CIP-B103-Lab8]
$ printf '<!DOCTYPE html>\n<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: Fuseini Imoru Kantuogaa</p></body></html>\n' | sudo tee /var/www/html/index.html
<!DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: Fuseini Imoru Kantuogaa</p></body></html>

(kali㉿kali)-[~/CIP-B103-Lab8]
$ curl http://127.0.0.1
<!DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: Fuseini Imoru Kantuogaa</p></body></html>

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sha256sum /var/www/html/index.html | tee reports/training_page_sha256.txt
```



```
a29eadb47a4a8a8475f6dab7e00a5955c1c86ccd758d2d68373a00f7527c2c2b /var/www/html/index.html
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ ip -br address | tee reports/interfaces.txt
```

```
lo                UNKNOWN          127.0.0.1/8 ::1/128
eth0              UP                192.168.44.128/24 fe80::fcdb:733e:9de1:56cb/64
docker0          DOWN              172.17.0.1/16
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ ip route | tee reports/routes.txt
```

```
default via 192.168.44.2 dev eth0 proto dhcp src 192.168.44.128 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.44.0/24 dev eth0 proto kernel scope link src 192.168.44.128 metric 100
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ ip neigh show | tee reports/arp_before.txt
```

```
192.168.44.2 dev eth0 lladdr 00:50:56:f4:fd:61 REACHABLE
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ ls -lh reports/
```

```
total 16K
-rw-rw-r-- 1 kali kali  58 Aug 10 19:00 arp_before.txt
-rw-rw-r-- 1 kali kali 180 Aug 10 18:59 interfaces.txt
-rw-rw-r-- 1 kali kali 231 Aug 10 19:00 routes.txt
-rw-rw-r-- 1 kali kali  91 Aug 10 18:59 training_page_sha256.txt
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ ip -br address
```

```
lo                UNKNOWN          127.0.0.1/8 ::1/128
eth0              UP                192.168.44.128/24 fe80::fcdb:733e:9de1:56cb/64
docker0          DOWN              172.17.0.1/16
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ ip route
```

```
default via 192.168.44.2 dev eth0 proto dhcp src 192.168.44.128 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.44.0/24 dev eth0 proto kernel scope link src 192.168.44.128 metric 100
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ ip neigh show
```

```
192.168.44.2 dev eth0 lladdr 00:50:56:f4:fd:61 REACHABLE
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ ip -br address
```

```
lo                UNKNOWN          127.0.0.1/8 ::1/128
eth0              UP                192.168.44.128/24 fe80::fcdb:733e:9de1:56cb/64
docker0          DOWN              172.17.0.1/16
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ ip neigh show
```

```

└─$ ip neigh show
192.168.44.2 dev eth0 lladdr 00:50:56:f4:fd:61 REACHABLE

(kali㉿kali)-[~/CIP-B103-Lab8]
└─$ resolvectl status
Global
    Protocols: +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 3 (docker0)
    Current Scopes: none
    Protocols: -DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Default Route: no

Link 2 (eth0)
    Current Scopes: DNS LLMNR/IPv4 LLMNR/IPv6
    Protocols: +DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
Current DNS Server: 192.168.44.2
    DNS Servers: 192.168.44.2
    DNS Domain: localdomain
    Default Route: yes

```

```

(kali㉿kali)-[~/CIP-B103-Lab8]
└─$ arping -I eth0 -c 3 192.168.44.1
arping: libnet_init(LIBNET_LINK, eth0): libnet_open_link(): UID/EUID 0 or capability CAP_NET_RAW required
arping: you may need to run as root

(kali㉿kali)-[~/CIP-B103-Lab8]
└─$ sudo arping -I eth0 -c 3 192.168.44.1
ARPING 192.168.44.1
60 bytes from 00:50:56:c0:00:08 (192.168.44.1): index=0 time=251.806 usec
60 bytes from 00:50:56:c0:00:08 (192.168.44.1): index=1 time=737.632 usec
60 bytes from 00:50:56:c0:00:08 (192.168.44.1): index=2 time=953.797 usec

— 192.168.44.1 statistics —
3 packets transmitted, 3 packets received, 0% unanswered (0 extra)
rtt min/avg/max/std-dev = 0.252/0.648/0.954/0.294 ms

```

```

(kali㉿kali)-[~/CIP-B103-Lab8]
└─$ sudo tshark -i eth0 \
-f 'host 192.168.44.1 and (arp or port 53 or tcp port 80)' \
-a duration:30 \
-w /tmp/dns_baseline.pcapng &
sleep 3
[1] 21189
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'

(kali㉿kali)-[~/CIP-B103-Lab8]
└─$ ls -lh /tmp/dns_baseline.pcapng
-rw-r--r-- 1 root root 2.0K Aug 10 19:35 /tmp/dns_baseline.pcapng

(kali㉿kali)-[~/CIP-B103-Lab8]
└─$ sudo cp /tmp/dns_baseline.pcapng evidence/dns_baseline.pcapng
sudo chown kali:kali evidence/dns_baseline.pcapng

```



```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ ls -lh evidence/dns_baseline.pcapng
-rw-r--r-- 1 kali kali 2.0K Aug 10 19:35 evidence/dns_baseline.pcapng

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sha256sum evidence/dns_baseline.pcapng | tee reports/dns_baseline_sha256.txt
290843ddb34d7271387b064cd782490fd389fcfefa10702a68ebc1f1b7828749 evidence/dns_baseline.pcapng

(kali㉿kali)-[~/CIP-B103-Lab8]
$ capinfos evidence/dns_baseline.pcapng
File name:          evidence/dns_baseline.pcapng
File type:          Wireshark/... - pcapng
File encapsulation: Ethernet
File timestamp precision: nanoseconds (9)
Packet size limit:  file hdr: (not set)
Number of packets:  18
File size:          2,044 bytes
Data size:          1,080 bytes
Capture duration:   24.128390513 seconds
Earliest packet time: 2026-08-10 19:34:30.911250478
Latest packet time:  2026-08-10 19:34:55.039640991
Data byte rate:     44 bytes/s
Data bit rate:      358 bits/s
```

```
Data bit rate:      358 bits/s
Average packet size: 60.00 bytes
Average packet rate: 0 packets/s
SHA256:             290843ddb34d7271387b064cd782490fd389fcfefa10702a68ebc1f1b7828749
SHA1:               76d8b31106de0498f231a5d0d8723a3e7cd582ed
Strict time order:  True
Capture hardware:   Intel(R) Core(TM) i5-7300HQ CPU @ 2.50GHz (with SSE4.2)
Capture oper-sys:   Linux 6.18.12+kali-amd64
Capture application: Dumpcap (Wireshark) 4.6.6
Number of interfaces in file: 1
Interface #0 info:
    Name = eth0
    Encapsulation = Ethernet (1 - ether)
    Capture length = 262144
    Time precision = nanoseconds (9)
    Time ticks per second = 1000000000
    Time resolution = 0x09
    Filter string = host 192.168.44.1 and (arp or port 53 or tcp port 80)
    Operating system = Linux 6.18.12+kali-amd64
    Number of stat entries = 1
    Number of packets = 18
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ tshark -r evidence/dns_baseline.pcapng -Y 'dns' \
-T fields \
-e frame.number \
-e frame.time \
-e ip.src \
-e ip.dst \
-e dns.qry.name \
-e dns.qry.type \
-e dns.a \
-e dns.resp.ttl

(kali㉿kali)-[~/CIP-B103-Lab8]
$ tshark -r evidence/dns_baseline.pcapng -Y arp
1 0.000000000 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
2 0.746548070 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
3 1.651379513 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
4 2.721333579 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
5 3.451266511 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
6 4.351598248 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
7 8.149989020 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
8 8.854731160 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
9 9.746901170 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
10 10.854441063 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
```

```

10 10.854441063 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
11 11.544560351 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
12 12.451323366 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
13 16.272774132 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
14 16.949290599 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
15 17.845899397 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
16 22.565820229 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
17 23.229560281 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1
18 24.128390513 VMware_c0:00:08 → Broadcast ARP 60 Who has 192.168.44.2? Tell 192.168.44.1

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo tshark -i eth0 \
-f 'host 192.168.44.1 and (arp or port 53 or tcp port 80)' \
-a duration:30 \
-w /tmp/dns_baseline2.pcapng &
[1] 22527

(kali㉿kali)-[~/CIP-B103-Lab8]
$ Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
17

[1] + done sudo tshark -i eth0 -f 'host 192.168.44.1 and (arp or port 53 or tcp port 80)'
(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo cp /tmp/dns_baseline2.pcapng evidence/dns_baseline.pcapng

```

```

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sysctl net.ipv4.ip_forward | tee reports/ip_forward_before.txt
net.ipv4.ip_forward = 1

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo iptables-save | tee reports/iptables_before.rules
# Generated by iptables-save v1.8.13 (nf_tables) on Mon Aug 10 19:45:42 2026
*filter
:INPUT ACCEPT [0:0]
:FORWARD DROP [0:0]
:OUTPUT ACCEPT [0:0]
:DOCKER - [0:0]
:DOCKER-BRIDGE - [0:0]
:DOCKER-CT - [0:0]
:DOCKER-FORWARD - [0:0]
:DOCKER-ISOLATION-STAGE-1 - [0:0]
:DOCKER-ISOLATION-STAGE-2 - [0:0]
:DOCKER-USER - [0:0]
-A FORWARD -j DOCKER-USER
-A FORWARD -j DOCKER-FORWARD
-A DOCKER ! -i docker0 -o docker0 -j DROP
-A DOCKER-BRIDGE -o docker0 -j DOCKER
-A DOCKER-CT -o docker0 -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A DOCKER-FORWARD -j DOCKER-CT

```

```
-A DOCKER-FORWARD -j DOCKER-CT
-A DOCKER-FORWARD -j DOCKER-ISOLATION-STAGE-1
-A DOCKER-FORWARD -j DOCKER-BRIDGE
-A DOCKER-FORWARD -i docker0 -j ACCEPT
-A DOCKER-ISOLATION-STAGE-1 -i docker0 ! -o docker0 -j DOCKER-ISOLATION-STAGE-2
-A DOCKER-ISOLATION-STAGE-2 -o docker0 -j DROP
COMMIT
```

```
# Completed on Mon Aug 10 19:45:42 2026
```

```
# Generated by iptables-save v1.8.13 (nf_tables) on Mon Aug 10 19:45:42 2026
```

```
*nat
```

```
:PREROUTING ACCEPT [21:1704]
```

```
:INPUT ACCEPT [0:0]
```

```
:OUTPUT ACCEPT [424:31411]
```

```
:POSTROUTING ACCEPT [424:31411]
```

```
:DOCKER - [0:0]
```

```
-A PREROUTING -m addrtype --dst-type LOCAL -j DOCKER
```

```
-A OUTPUT ! -d 127.0.0.0/8 -m addrtype --dst-type LOCAL -j DOCKER
```

```
-A POSTROUTING -s 172.17.0.0/16 ! -o docker0 -j MASQUERADE
```

```
-A DOCKER -i docker0 -j RETURN
```

```
COMMIT
```

```
# Completed on Mon Aug 10 19:45:42 2026
```

```
(kali@kali)-[~/CIP-B103-Lab8]
```

```
$ sudo systemctl status apache2 --no-pager
```

```
● apache2.service - The Apache HTTP Server
```

```
● apache2.service - The Apache HTTP Server
```

```
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
```

```
Active: active (running) since Mon 2026-08-10 18:53:23 EDT; 52min ago
```

```
Invocation: 2a1cbd563ff246c6a843f21a71f0836b
```

```
Docs: https://httpd.apache.org/docs/2.4/
```

```
Main PID: 892 (apache2)
```

```
Status: "Total requests: 2; Idle/Busy workers 100/0; Requests/sec: 0.00063; Bytes served/sec: 3 B/sec"
```

```
Tasks: 6 (limit: 2075)
```

```
Memory: 12.7M (peak: 36.7M, swap: 8.9M, swap peak: 8.9M)
```

```
CPU: 922ms
```

```
CGroup: /system.slice/apache2.service
```

```
└─892 /usr/sbin/apache2 -k start -DFOREGROUND
```

```
└─950 /usr/sbin/apache2 -k start -DFOREGROUND
```

```
└─951 /usr/sbin/apache2 -k start -DFOREGROUND
```

```
└─952 /usr/sbin/apache2 -k start -DFOREGROUND
```

```
└─953 /usr/sbin/apache2 -k start -DFOREGROUND
```

```
└─954 /usr/sbin/apache2 -k start -DFOREGROUND
```

```
Aug 10 18:53:21 kali systemd[1]: Starting apache2.service - The Apache HTTP Server...
```

```
Aug 10 18:53:23 kali apachectl[892]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
```

```
Aug 10 18:53:23 kali systemd[1]: Started apache2.service - The Apache HTTP Server.
```

```
Hint: Some lines were ellipsized, use -l to show in full.
```

```
└─$ apache2ctl -S
```

```
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
```

```
VirtualHost configuration:
```

```
*:80 127.0.1.1 (/etc/apache2/sites-enabled/000-default.conf:1)
```

```
ServerRoot: "/etc/apache2"
```

```
Main DocumentRoot: "/var/www/html"
```

```
Main ErrorLog: "/var/log/apache2/error.log"
```

```
Mutex default: dir="/run/apache2/" mechanism=default
```

```
Mutex mpm-accept: using_defaults
```

```
Mutex watchdog-callback: using_defaults
```

```
PidFile: "/run/apache2/apache2.pid"
```

```
Define: DUMP_VHOSTS
```

```
Define: DUMP_RUN_CFG
```

```
User: name="www-data" id=33 not used
```

```
Group: name="www-data" id=33 not used
```

```
(kali@kali)-[~/CIP-B103-Lab8]
```

```
$ grep -R "DocumentRoot" /etc/apache2/sites-enabled/ /etc/apache2/sites-available/ 2>/dev/null
```

```
/etc/apache2/sites-enabled/000-default.conf: DocumentRoot /var/www/html
```

```
/etc/apache2/sites-available/000-default.conf: DocumentRoot /var/www/html
```

```
/etc/apache2/sites-available/default-ssl.conf: DocumentRoot /var/www/html
```

```
(kali@kali)-[~/CIP-B103-Lab8]
```

```
$ sudo -u www-data cat /var/www/html/index.html
```





```

└─$ sudo tail -20 /var/log/apache2/access.log
127.0.0.1 - - [26/Jul/2026:11:25:24 -0400] "GET / HTTP/1.1" 200 10958 "-" "curl/8.20.0"
127.0.0.1 - - [28/Jul/2026:07:50:40 -0400] "GET /basic.html HTTP/1.1" 200 371 "-" "curl/8.20.0"
127.0.0.1 - - [28/Jul/2026:07:55:40 -0400] "GET /basic.html HTTP/1.1" 200 371 "-" "curl/8.20.0"
127.0.0.1 - - [28/Jul/2026:08:10:18 -0400] "GET /basic.html HTTP/1.1" 200 371 "-" "curl/8.20.0"
127.0.0.1 - - [28/Jul/2026:08:23:06 -0400] "GET /basic.html HTTP/1.1" 200 371 "-" "curl/8.20.0"
192.168.44.128 - - [01/Aug/2026:11:22:48 -0400] "GET /firewall_lab.html HTTP/1.1" 200 377 "-" "curl/8.20.0"
192.168.44.128 - - [01/Aug/2026:11:26:16 -0400] "GET /firewall_lab.html HTTP/1.1" 200 377 "-" "curl/8.20.0"
192.168.44.128 - - [01/Aug/2026:11:29:11 -0400] "GET /firewall_lab.html HTTP/1.1" 200 377 "-" "curl/8.20.0"
192.168.44.128 - - [01/Aug/2026:11:51:04 -0400] "GET /firewall_lab.html HTTP/1.1" 200 377 "-" "curl/8.20.0"
192.168.44.129 - - [01/Aug/2026:11:56:14 -0400] "GET /firewall_lab.html HTTP/1.1" 200 377 "-" "curl/8.20.0"
192.168.44.129 - - [01/Aug/2026:12:08:17 -0400] "GET /firewall_lab.html HTTP/1.1" 200 377 "-" "curl/8.20.0"
127.0.0.1 - - [10/Aug/2026:18:56:20 -0400] "GET / HTTP/1.1" 200 10958 "-" "curl/8.20.0"
127.0.0.1 - - [10/Aug/2026:18:59:28 -0400] "GET / HTTP/1.1" 200 466 "-" "curl/8.20.0"
192.168.44.128 - - [10/Aug/2026:19:46:40 -0400] "GET / HTTP/1.1" 200 466 "-" "curl/8.20.0"
192.168.44.1 - - [10/Aug/2026:19:46:59 -0400] "GET / HTTP/1.1" 200 466 "-" "curl/8.21.0"
192.168.44.128 - - [10/Aug/2026:19:52:38 -0400] "GET / HTTP/1.1" 200 466 "-" "curl/8.20.0"
192.168.44.128 - - [10/Aug/2026:19:55:28 -0400] "GET / HTTP/1.1" 200 456 "-" "curl/8.20.0"
192.168.44.1 - - [10/Aug/2026:19:55:56 -0400] "GET / HTTP/1.1" 200 456 "-" "curl/8.21.0"
127.0.0.1 - - [10/Aug/2026:19:57:32 -0400] "GET / HTTP/1.1" 200 456 "-" "curl/8.20.0"
192.168.44.128 - - [10/Aug/2026:19:57:56 -0400] "GET / HTTP/1.1" 200 456 "-" "curl/8.20.0"

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo tail -20 /var/log/apache2/error.log
mprotect() failed [13] Permission denied

```

```

mprotect() failed [13] Permission denied
[Mon Aug 10 08:57:41.080170 2026] [mpm_prefork:notice] [pid 918:tid 918] AH00163: Apache/2.4.68 (Debian) configur
red -- resuming normal operations
[Mon Aug 10 08:57:41.080487 2026] [core:notice] [pid 918:tid 918] AH00094: Command line: '/usr/sbin/apache2 -D F
OREGROUND'
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set th
e 'ServerName' directive globally to suppress this message
mprotect() failed [13] Permission denied
mprotect() failed [13] Permission denied
mprotect() failed [13] Permission denied
mprotect() failed [13] Permission denied
[Mon Aug 10 12:09:26.191879 2026] [mpm_prefork:notice] [pid 960:tid 960] AH00163: Apache/2.4.68 (Debian) configur
red -- resuming normal operations
[Mon Aug 10 12:09:26.192092 2026] [core:notice] [pid 960:tid 960] AH00094: Command line: '/usr/sbin/apache2 -D F
OREGROUND'
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set th
e 'ServerName' directive globally to suppress this message
mprotect() failed [13] Permission denied
mprotect() failed [13] Permission denied
mprotect() failed [13] Permission denied
mprotect() failed [13] Permission denied
[Mon Aug 10 18:53:23.826923 2026] [mpm_prefork:notice] [pid 892:tid 892] AH00163: Apache/2.4.68 (Debian) configur
red -- resuming normal operations
[Mon Aug 10 18:53:23.827148 2026] [core:notice] [pid 892:tid 892] AH00094: Command line: '/usr/sbin/apache2 -D F
OREGROUND'

```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo iptables-save | tee reports/iptables_before.rules
# Generated by iptables-save v1.8.13 (nf_tables) on Mon Aug 10 20:00:45 2026
*filter
:INPUT ACCEPT [0:0]
:FORWARD DROP [0:0]
:OUTPUT ACCEPT [0:0]
:DOCKER - [0:0]
:DOCKER-BRIDGE - [0:0]
:DOCKER-CT - [0:0]
:DOCKER-FORWARD - [0:0]
:DOCKER-ISOLATION-STAGE-1 - [0:0]
:DOCKER-ISOLATION-STAGE-2 - [0:0]
:DOCKER-USER - [0:0]
-A FORWARD -j DOCKER-USER
-A FORWARD -j DOCKER-FORWARD
-A DOCKER ! -i docker0 -o docker0 -j DROP
-A DOCKER-BRIDGE -o docker0 -j DOCKER
-A DOCKER-CT -o docker0 -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
-A DOCKER-FORWARD -j DOCKER-CT
-A DOCKER-FORWARD -j DOCKER-ISOLATION-STAGE-1
-A DOCKER-FORWARD -j DOCKER-BRIDGE
-A DOCKER-FORWARD -i docker0 -j ACCEPT
-A DOCKER-ISOLATION-STAGE-1 -i docker0 ! -o docker0 -j DOCKER-ISOLATION-STAGE-2
-A DOCKER-ISOLATION-STAGE-2 -o docker0 -j DROP
```

```
-A DOCKER-ISOLATION-STAGE-2 -o docker0 -j DROP
COMMIT
# Completed on Mon Aug 10 20:00:45 2026
# Generated by iptables-save v1.8.13 (nf_tables) on Mon Aug 10 20:00:45 2026
*nat
:PREROUTING ACCEPT [25:1908]
:INPUT ACCEPT [0:0]
:OUTPUT ACCEPT [455:33921]
:POSTROUTING ACCEPT [455:33921]
:DOCKER - [0:0]
-A PREROUTING -m addrtype --dst-type LOCAL -j DOCKER
-A OUTPUT ! -d 127.0.0.0/8 -m addrtype --dst-type LOCAL -j DOCKER
-A POSTROUTING -s 172.17.0.0/16 ! -o docker0 -j MASQUERADE
-A DOCKER -i docker0 -j RETURN
COMMIT
# Completed on Mon Aug 10 20:00:45 2026
```



```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ sed -n '1,240p' scripts/arp.py | tee reports/arp_script_review.txt
#!/usr/bin/env python3
```

```
import sys
import time

if len(sys.argv) != 3:
    print(f"Usage: sudo python3 {sys.argv[0]} VICTIM_IP GATEWAY_IP")
    sys.exit(1)

victim = sys.argv[1]
gateway = sys.argv[2]

print("=== ICDFA Authorized ARP Simulation ===")
print(f"Victim : {victim}")
print(f"Gateway: {gateway}")
print()
print("SIMULATION ONLY")
print("No ARP packets are modified or transmitted.")
print("Recording the intended ARP-spoofing event for the lab.")
print()

try:
    for i in range(30):
```

```
        for i in range(30):
            print(
                f"[{i+1:02d}/30] "
                f"Simulated ARP observation: victim={victim} gateway={gateway}"
            )
            time.sleep(1)
except KeyboardInterrupt:
    print("\nSimulation stopped.")
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ sed -n '1,260p' scripts/dns_spoof.py | tee reports/dns_script_review.txt
#!/usr/bin/env python3
```

```
import time

TRAINING_DOMAIN = "portal.icdfa.test"
REPLACEMENT_IP = "192.168.44.128"

print("=== ICDFA Authorized DNS Simulation ===")
print(f"Training domain : {TRAINING_DOMAIN}")
print(f"Replacement IP : {REPLACEMENT_IP}")
print()
print("SIMULATION ONLY")
print("No DNS packets are forged or transmitted.")
print("Recording the intended DNS-spoofing event for the lab.")
```

```

print()

try:
    for i in range(30):
        print(
            f"[{i+1:02d}/30] "
            f"Simulated DNS response: "
            f"{TRAINING_DOMAIN} → {REPLACEMENT_IP}"
        )
        time.sleep(1)
except KeyboardInterrupt:
    print("\nSimulation stopped.")

```

```

(kali㉿kali)-[~/CIP-B103-Lab8]
$ ls -lh scripts/
total 8.0K
-rwxrwxr-x 1 kali kali 710 Aug 10 20:17 arp.py
-rwxrwxr-x 1 kali kali 668 Aug 10 20:17 dns_spoof.py

```

```

(kali㉿kali)-[~/CIP-B103-Lab8]
$

```

```

C:\Users\user>curl http://192.168.44.128/
<!DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Training Page</h1><p>This is an authorized simulation. Do not enter credentials.</p><p>Analyst: Fuseini Imoru</p></body></html>
C:\Users\user>

```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo tshark -i eth0 \
-f 'host 192.168.44.1 and (arp or port 53 or tcp port 80)' \
-a duration:60 \
-w /tmp/dns_spoof_controlled.pcapng &
[1] 42845

(kali㉿kali)-[~/CIP-B103-Lab8]
$ Running as user "root" and group "root". This could be dange
Capturing on 'eth0'
16

[1] + done      sudo tshark -i eth0 -f 'host 192.168.44.1 and
(kali㉿kali)-[~/CIP-B103-Lab8]
$ ls -l scripts/arp.py scripts/dns_spoof.py
-rwxrwxr-x 1 kali kali 710 Aug 10 20:17 scripts/arp.py
-rwxrwxr-x 1 kali kali 668 Aug 10 20:17 scripts/dns_spoof.py

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sed -n '1,240p' scripts/arp.py
#!/usr/bin/env python3

import sys
import time

if len(sys.argv) != 3:
    print(f"Usage: sudo python3 {sys.argv[0]} VICTIM_IP GATEWAY_
    sys.exit(1)

victim = sys.argv[1]
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
└─$ sed -n '1,260p' scripts/dns_spoof.py
#!/usr/bin/env python3

import time

TRAINING_DOMAIN = "portal.icdfa.test"
REPLACEMENT_IP = "192.168.44.128"

print("≡≡≡ ICDFA Authorized DNS Simulation ≡≡≡")
print(f"Training domain : {TRAINING_DOMAIN}")
print(f"Replacement IP   : {REPLACEMENT_IP}")
print()
print("SIMULATION ONLY")
print("No DNS packets are forged or transmitted.")
print("Recording the intended DNS-spoofing event for the lab.")
print()

try:
    for i in range(30):
        print(
            f"[{i+1:02d}/30] "
            f"Simulated DNS response: "
            f"{TRAINING_DOMAIN} → {REPLACEMENT_IP}"
        )
        time.sleep(1)
except KeyboardInterrupt:
    print("\nSimulation stopped.")
```



```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ curl -v http://192.168.44.128/
* Trying 192.168.44.128:80 ...
* Established connection to 192.168.44.128 (192.168.44.128 port
* using HTTP/1.x
> GET / HTTP/1.1
> Host: 192.168.44.128
> User-Agent: curl/8.20.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Date: Tue, 11 Aug 2026 00:28:21 GMT
< Server: Apache/2.4.68 (Debian)
< Last-Modified: Mon, 10 Aug 2026 23:54:26 GMT
< ETag: "cd-658ba1306a4aa"
< Accept-Ranges: bytes
< Content-Length: 205
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Tra
on. Do not enter credentials.</p><p>Analyst: Fuseini Imoru</p></
* Connection #0 to host 192.168.44.128:80 left intact

(kali㉿kali)-[~/CIP-B103-Lab8]
$ cat /var/www/html/index.html
<!DOCTYPE html>
<html><body style="font-family:Arial"><h1>ICDFA DNS Spoofing Tra
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo tshark -i eth0 \
-f 'host 192.168.44.1 and (arp or port 53 or tcp port 80)' \
-a duration:60 \
-w /tmp/dns_spoof_controlled.pcapng
Running as user "root" and group "root". This could be dangerous
Capturing on 'eth0'
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo cp /tmp/dns_spoof_controlled.pcapng evidence/
sudo chown kali:kali evidence/dns_spoof_controlled.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ ls -lh evidence/dns_spoof_controlled.pcapng
-rw-r--r-- 1 kali kali 388 Aug 10 20:30 evidence/dns_spoof_contr
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ sha256sum evidence/dns_spoof_controlled.pcapng \
| tee reports/dns_spoof_controlled_sha256.txt
6d06affbbcbdddc10d91abf4071a9e25930bb3ee81509d7f916bb786441a12ff
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
$
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]  
$ sudo timeout 30 python3 scripts/arp.py 192.168.44.1 192.168.44.2  
[sudo] password for kali:  
≡ ICDFA Authorized ARP Simulation ≡  
Victim : 192.168.44.1  
Gateway: 192.168.44.2  
  
SIMULATION ONLY  
No ARP packets are modified or transmitted.  
Recording the intended ARP-spoofing event for the lab.  
  
[01/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[02/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[03/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[04/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[05/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[06/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[07/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2
```

```
[08/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[09/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[10/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[11/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[12/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[13/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[14/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[15/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[16/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[17/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[18/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[19/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[20/30] Simulated ARP observation: victim=192.168.44.1 gate  
way=192.168.44.2  
[21/30] Simulated ARP observation: victim=192.168.44.1 gate
```



```
way=192.168.44.2
[24/30] Simulated ARP observation: victim=192.168.44.1 gate
way=192.168.44.2
[25/30] Simulated ARP observation: victim=192.168.44.1 gate
way=192.168.44.2
[26/30] Simulated ARP observation: victim=192.168.44.1 gate
way=192.168.44.2
[27/30] Simulated ARP observation: victim=192.168.44.1 gate
way=192.168.44.2
[28/30] Simulated ARP observation: victim=192.168.44.1 gate
way=192.168.44.2
[29/30] Simulated ARP observation: victim=192.168.44.1 gate
way=192.168.44.2
[30/30] Simulated ARP observation: victim=192.168.44.1 gate
way=192.168.44.2
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo timeout 30 python3 scripts/dns_spoof.py
```

```
≡ ICDFA Authorized DNS Simulation ≡
```

```
Training domain : portal.icdfa.test
```

```
Replacement IP : 192.168.44.128
```

SIMULATION ONLY

No DNS packets are forged or transmitted.

Recording the intended DNS-spoofing event for the lab.

```
[01/30] Simulated DNS response: portal.icdfa.test → 192.16
```

Recording the intended DNS-spoofing event for the lab.

[01/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

[02/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

[03/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

[04/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

[05/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

[06/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

[07/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

[08/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

[09/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

[10/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

[11/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

[12/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

[13/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128

```
8.44.128
[15/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
[16/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
[17/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
[18/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
[19/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
[20/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
[21/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
[22/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
[23/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
[24/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
[25/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
[26/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
[27/30] Simulated DNS response: portal.icdfa.test → 192.16
8.44.128
```

```
[28/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128
[29/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128
[30/30] Simulated DNS response: portal.icdfa.test → 192.168.44.128
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ ls -lh /tmp/dns_spoof_controlled.pcapng
```

```
-rw-r--r-- 1 root root 388 Aug 10 20:30 /tmp/dns_spoof_controlled.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ sudo cp /tmp/dns_spoof_controlled.pcapng evidence/dns_spoof_controlled.pcapng
```

```
sudo chown kali:kali evidence/dns_spoof_controlled.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ ls -lh evidence/dns_spoof_controlled.pcapng
```

```
-rw-r--r-- 1 kali kali 388 Aug 10 20:35 evidence/dns_spoof_controlled.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ sha256sum evidence/dns_spoof_controlled.pcapng | tee reports/dns_spoof_controlled_sha256.txt
```

```
6d06affbbcbddc10d91abf4071a9e25930bb3ee81509d7f916bb786441a12ff evidence/dns_spoof_controlled.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ sha256sum evidence/dns_spoof_controlled.pcapng | tee reports/dns_spoof_controlled_sha256.txt
```

```
6d06affbbcbddc10d91abf4071a9e25930bb3ee81509d7f916bb786441a12ff evidence/dns_spoof_controlled.pcapng
```

```
(kali㉿kali)-[~/CIP-B103-Lab8]
```

```
$ tshark -r evidence/dns_spoof_controlled.pcapng -q -z io,phs
```

---

---

```
Protocol Hierarchy Statistics
Filter:
```

---

---

```

(kali㉿kali)-[~/CIP-B103-Lab8]
$ PCAP=evidence/dns_spoof_controlled.pcapng

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sha256sum "$PCAP" | tee reports/dns_spoof_capture_sha256.txt
6d06affbbcbddc10d91abf4071a9e25930bb3ee81509d7f916bb786441a12ff  evidence/dns_spoof_controlled.pcapng

(kali㉿kali)-[~/CIP-B103-Lab8]
$ tshark -r "$PCAP" -Y 'dns' -T fields -e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e
udp.srcport -e ip.dst -e udp.dstport -e dns.id -e dns.flags.response -e dns.qry.name -e dns.qry.type -e dns.flag
s.rcode -e dns.a -e dns.resp.ttl | tee reports/dns_all_fields.tsv

(kali㉿kali)-[~/CIP-B103-Lab8]
$ wc -l reports/dns_all_fields.tsv
0 reports/dns_all_fields.tsv

(kali㉿kali)-[~/CIP-B103-Lab8]
$ tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields -e frame.number -e frame.time_epoch -e arp.src.proto_ipv4 -e
arp.src.hw_mac -e arp.dst.proto_ipv4 -e arp.dst.hw_mac | tee reports/arp_claims_during_spoof.tsv

(kali㉿kali)-[~/CIP-B103-Lab8]
$ wc -l reports/arp_claims_during_spoof.tsv
0 reports/arp_claims_during_spoof.tsv

```

```

(kali㉿kali)-[~/CIP-B103-Lab8]
$ tshark -r "$PCAP" -Y 'http.request || tcp.flags.syn==1' -T fields -e frame.number -e frame.time_epoch -e ip.
src -e tcp.srcport -e ip.dst -e tcp.dstport -e http.host -e http.request.uri | tee reports/post_dns_connections.
tsv

(kali㉿kali)-[~/CIP-B103-Lab8]
$ wc -l reports/post_dns_connections.tsv
0 reports/post_dns_connections.tsv

(kali㉿kali)-[~/CIP-B103-Lab8]
$ ls -lh reports/dns_spoof_capture_sha256.txt \
reports/dns_all_fields.tsv \
reports/arp_claims_during_spoof.tsv \
reports/post_dns_connections.tsv
-rw-rw-r-- 1 kali kali  0 Aug 10 20:46 reports/arp_claims_during_spoof.tsv
-rw-rw-r-- 1 kali kali  0 Aug 10 20:46 reports/dns_all_fields.tsv
-rw-rw-r-- 1 kali kali 103 Aug 10 20:45 reports/dns_spoof_capture_sha256.txt
-rw-rw-r-- 1 kali kali  0 Aug 10 20:47 reports/post_dns_connections.tsv

(kali㉿kali)-[~/CIP-B103-Lab8]
$ cat reports/dns_spoof_capture_sha256.txt
6d06affbbcbddc10d91abf4071a9e25930bb3ee81509d7f916bb786441a12ff  evidence/dns_spoof_controlled.pcapng

(kali㉿kali)-[~/CIP-B103-Lab8]
$

```

```

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo pkill -f 'arp.py' 2>/dev/null || true
sudo pkill -f 'dns_spoof.py' 2>/dev/null || true
[sudo] password for kali:

(kali㉿kali)-[~/CIP-B103-Lab8]
$ ps aux | grep -E '[a]rp.py|[d]ns_spoof.py' | tee reports/process_cleanup_check.txt

(kali㉿kali)-[~/CIP-B103-Lab8]
$ cat reports/ip_forward_before.txt
net.ipv4.ip_forward = 1

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sysctl net.ipv4.ip_forward | tee reports/ip_forward_after_cleanup.txt
net.ipv4.ip_forward = 1

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo iptables -L -n -v --line-numbers | tee reports/iptables_after_cleanup.txt
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                 destination

Chain FORWARD (policy DROP 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                 destination
1      0      0 DOCKER-USER  all  --  *      *           0.0.0.0/0             0.0.0.0/0
2      0      0 DOCKER-FORWARD  all  --  *      *           0.0.0.0/0             0.0.0.0/0

```



```
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
num  pkts bytes target    prot opt in     out     source                 destination

Chain DOCKER (1 references)
num  pkts bytes target    prot opt in     out     source                 destination
1    0    0 DROP      all  --  !docker0 docker0  0.0.0.0/0             0.0.0.0/0

Chain DOCKER-BRIDGE (1 references)
num  pkts bytes target    prot opt in     out     source                 destination
1    0    0 DOCKER    all  --  *      docker0  0.0.0.0/0             0.0.0.0/0

Chain DOCKER-CT (1 references)
num  pkts bytes target    prot opt in     out     source                 destination
1    0    0 ACCEPT   all  --  *      docker0  0.0.0.0/0             0.0.0.0/0      ctstate RELATED,E
STABLISHED

Chain DOCKER-FORWARD (1 references)
num  pkts bytes target    prot opt in     out     source                 destination
1    0    0 DOCKER-CT all  --  *      *        0.0.0.0/0             0.0.0.0/0
2    0    0 DOCKER-ISOLATION-STAGE-1 all  --  *      *        0.0.0.0/0             0.0.0.0/0
3    0    0 DOCKER-BRIDGE all  --  *      *        0.0.0.0/0             0.0.0.0/0
4    0    0 ACCEPT   all  --  docker0 *        0.0.0.0/0             0.0.0.0/0

Chain DOCKER-ISOLATION-STAGE-1 (1 references)
num  pkts bytes target    prot opt in     out     source                 destination
1    0    0 DOCKER-ISOLATION-STAGE-2 all  --  docker0 !docker0 0.0.0.0/0             0.0.0.0/0
```

```
Chain DOCKER-ISOLATION-STAGE-2 (1 references)
num  pkts bytes target    prot opt in     out     source                 destination
1    0    0 DROP      all  --  *      docker0  0.0.0.0/0             0.0.0.0/0

Chain DOCKER-USER (1 references)
num  pkts bytes target    prot opt in     out     source                 destination

(kali㉿kali)-[~/CIP-B103-Lab8]
$ sudo ip neigh flush all

(kali㉿kali)-[~/CIP-B103-Lab8]
$ ip neigh show | tee reports/arp_after_cleanup.txt

(kali㉿kali)-[~/CIP-B103-Lab8]
$ ps aux | grep -E '[a]rp.py|[d]ns_spoof.py' | tee reports/process_cleanup_check.txt

(kali㉿kali)-[~/CIP-B103-Lab8]
$ echo "=== IP forwarding ==="
sysctl net.ipv4.ip_forward

echo
echo "=== Running simulation scripts ==="
ps aux | grep -E '[a]rp.py|[d]ns_spoof.py' || true
```

```
echo
echo "=== Evidence files ==="
ls -lh evidence/

echo
echo "=== Cleanup reports ==="
ls -lh reports/*cleanup* reports/arp_after_cleanup.txt 2>/dev/null
=== IP forwarding ===
net.ipv4.ip_forward = 1

=== Running simulation scripts ===

=== Evidence files ===
total 8.0K
-rw-r--r-- 1 kali kali 2.0K Aug 10 19:38 dns_baseline.pcapng
-rw-r--r-- 1 kali kali 388 Aug 10 20:35 dns_spoof_controlled.pcapng

=== Cleanup reports ===
-rw-rw-r-- 1 kali kali 0 Aug 10 20:52 reports/arp_after_cleanup.txt
-rw-rw-r-- 1 kali kali 0 Aug 10 20:52 reports/arp_after_cleanup.txt
-rw-rw-r-- 1 kali kali 24 Aug 10 20:51 reports/ip_forward_after_cleanup.txt
-rw-rw-r-- 1 kali kali 2.5K Aug 10 20:51 reports/iptables_after_cleanup.txt
-rw-rw-r-- 1 kali kali 0 Aug 10 20:52 reports/process_cleanup_check.txt

(kali@kali)-[~/CIP-B103-Lab8]
$
```